

UNIVERSITY OF BAHRAIN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER 2024 -2025

TEST 2 - Key

MATHS. 210

DATE: 25 - 11 - 2024

TIME: 1:00 PM - 2:00 PM

EXAM INSTRUCTIONS:

- 1) Number of pages is 5 including the cover page.
 - 2) Show all your work, except for question one.
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NAME :	I.D. NUMBER:	SERIAL NUMBER:

<i>Q.1</i>	<i>Q.2</i>	<i>Q.3</i>	<i>Q.4</i>	<i>OUT OF</i>
<i>16</i>	<i>8</i>	<i>8</i>	<i>8</i>	<i>40</i>

GOOD LUCK

QUESTION 1 (2 marks for each question)

Fill in the blanks to make a correct statement:

1) Write a condition about k so that the matrix $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & k & 2 \end{pmatrix}$ is invertible

$$k \neq 1$$

2) If A is a matrix of size 2×2 and $\det(A) = 3$, then

$$\det(3 A A^T A^{-1}) = 27$$

3) If $H = \{ (2x, x + y, x - y, a^2 - 1) : x, y \in \mathbb{R} \}$ is a subspace of $\mathbb{R}^4$, then

$$a = \pm 1$$

4) Suppose that $(u)_S = (1, 2, -2)$ where $S = \{ (1, -1, 0), (2, 0, 1), (0, 1, 3) \}$ is a basis of $\mathbb{R}^3$, then

$$u = (5, -3, -4)$$

5) If $S = \{v_1, v_2, v_3, v_4\}$ spans $(\mathbb{R}^n)$, then $n \leq 4$ or $n \geq 4$ or $n = 4$. Which statement is correct?

$$n \leq 4$$

6) If $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 3 & 3 & 3 \end{pmatrix}$, then the rank of A is

$$2$$

7) If A is a matrix of size 6×4 and $\text{nullity}(A^T) = 3$ $\text{nullity}(A)$, then

$$\text{nullity}(A) = 1$$

8) If $\begin{vmatrix} 1 & 1 & 1 \\ 2 & x & 1 \\ 3 & x^2 & 1 \end{vmatrix} = 0$, then

$$x = 1$$

QUESTION 2 (3 + 3 + 2 marks)

(Show your work)

a) Find a condition about t so that that $B = \{v_1, v_2, v_3\}$ is a basis of $\mathbb{R}^3$, where

$$v_1 = (0, t-1, 0) \quad , \quad v_2 = (0, 3, 1) \quad , \quad v_3 = (1, 0, 3)$$

Solution:Since B has 3 vectors and $\dim \mathbb{R}^3 = 3$, it is sufficient to examine when B is linearly independent.

$$av_1 + bv_2 + cv_3 = (0, 0, 0) \quad \Rightarrow \quad a(0, t-1, 0) + b(0, 3, 1) + c(1, 0, 3) = (0, 0, 0) \quad (1)$$

$$\Rightarrow \quad (c, a(t-1) + 3b, b + 3c) = (0, 0, 0)$$

$$\Rightarrow \quad c = 0$$

$$a(t-1) + 3b = 0$$

$$b + 3c = 0 \quad (1)$$

$$\Delta = \begin{vmatrix} 0 & 0 & 1 \\ t-1 & 3 & 0 \\ 0 & 1 & 3 \end{vmatrix} = \begin{vmatrix} t-1 & 3 \\ 0 & 1 \end{vmatrix} = t-1.$$

(1)

$$B \text{ is a basis of } \mathbb{R}^3 \Leftrightarrow B \text{ is linearly independent} \Leftrightarrow \Delta \neq 0 \Leftrightarrow t \neq 1$$

b) Suppose that $t = 2$. Find the coordinates of $v = (1, -1, 3)$ relative to B .**Solution:**

$$av_1 + bv_2 + cv_3 = v \quad \Rightarrow \quad a(0, 1, 0) + b(0, 3, 1) + c(1, 0, 3) = (1, -1, 3) \quad (1)$$

$$\Rightarrow \quad (c, a + 3b, b + 3c) = (1, -1, 3)$$

$$\Rightarrow \quad c = 1$$

$$a + 3b = -1$$

$$b + 3c = 3$$

$$\Rightarrow \quad c = 1; \quad b = 0, \quad a = -1 \quad (1)$$

$$\Rightarrow \quad (v)_S = (-1, 0, 1) \quad (1)$$

c) Is $S = \{v_1, v_2, v_3, v\}$ linearly independent? Explain.**Solution:**Since $\text{card}(S) = 4 > \dim \mathbb{R}^3 = 3$, then S is linearly dependent. (1)

QUESTION 3 (4 + 4 marks)

Let $H = \{ (x + 2y, x - y, -3x) : x, y \in \mathbb{R} \}$.

a) Prove that H is a subspace of $\mathbb{R}^3$.

Solution:

$$\begin{aligned} H &= \{ (x + 2y, x - y, -3x) : x, y \in \mathbb{R} \} \\ &= \{ x(1, 1, -3) + y(2, -1, 0) : x, y \in \mathbb{R} \} \end{aligned}$$

Set $u = (1, 1, -3)$ and $v = (2, -1, 0)$, then $\{u, v\}$ is a spanning set of H , and H is a subspace of $\mathbb{R}^3$.

b) Find $\dim(H)$.

Solution:

$\{u, v\}$ is a spanning set of H . To show that $\{u, v\}$ is a basis of H , it remains to show that $\{u, v\}$ is linearly independent:

$$\alpha(1, 1, -3) + \beta(2, -1, 0) = (0, 0, 0)$$

implies

$$(\alpha + 2\beta, \alpha - \beta, -3\alpha) = (0, 0, 0), \text{ so } \alpha = \beta = 0.$$

Hence, $\{u, v\}$ is a basis of H and $\dim H = 2$.

QUESTION 4 (4 + 4 marks)

Let $A' = \begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ be a row-echelon form of A .

a) Find a basis of the nullspace of A .

Solution:

Let us transform the matrix A' to a reduced row-echelon form

$$A' = \begin{bmatrix} \mathbf{1} & 3 & 0 & 2 \\ 0 & \mathbf{1} & 1 & 1 \\ 0 & 0 & 0 & \mathbf{1} \end{bmatrix} \sim \begin{bmatrix} \mathbf{1} & 3 & 0 & 2 \\ 0 & \mathbf{1} & 1 & 0 \\ 0 & 0 & 0 & \mathbf{1} \end{bmatrix} \sim \begin{bmatrix} \mathbf{1} & 3 & 0 & 0 \\ 0 & \mathbf{1} & 1 & 0 \\ 0 & 0 & 0 & \mathbf{1} \end{bmatrix} \sim \begin{bmatrix} \mathbf{1} & 0 & -3 & 0 \\ 0 & \mathbf{1} & 1 & 0 \\ 0 & 0 & 0 & \mathbf{1} \end{bmatrix} = A''$$

$$\begin{bmatrix} \mathbf{1} & 0 & -3 & 0 \\ 0 & \mathbf{1} & 1 & 0 \\ 0 & 0 & 0 & \mathbf{1} \end{bmatrix} X = 0 \Rightarrow x_1 - 3x_3 = 0 ; x_2 + x_3 = 0 ; x_4 = 0$$

Let $x_3 = r$ be a free variable, then the leading variables are $x_1 = 3r$, $x_2 = -r$ and $x_4 = 0$.

Thus $N(A) = \{ (3r, -r, r, 0) : r \text{ is real number} \} = \text{Span}\{(3, -1, 1, 0)\}$.

Since $\dim N(A) = \text{Nullity}(A) = 4 - 3 = 1$, then $\{(3, -1, 1, 0)\}$ is a basis of $N(A)$.

b) If c_1, c_2, c_3, c_4 are the column vectors of A , write c_3 as a linear combination of c_1, c_2, c_4 .

Solution:

Let c_1', c_2', c_3', c_4' be the column vectors of A'' .

Then $c_3' = -3c_1' + c_2'$. It follows that $c_3 = -3c_1 + c_2$.